Architecture Selection & Justification

Micro-Lending & Peer Credit Risk Assessor — Lab 3: Component Modelling & Architectural Pattern Selection

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We chose Layered Architecture for the Micro-Lending & Peer Credit Risk Assessor System.

Architectural Choice

The system is structured into three horizontal layers — a Presentation Layer (User Interface Component), a Business Layer (Risk Assessment, Loan Management, Transaction Processing, and Notification components), and a Data Layer (Database Component). Each layer depends only on the layer directly below it, and components communicate through well-defined provided/required interfaces.

Reason 1 — Transactional Integrity for Financial Operations

NFR-001 requires that loan repayment transactions and EMI distribution execute inside ACID-compliant database transactions to prevent double-spending. A Layered Architecture centralises all persistence behind a single Database component in the Data Layer, so the Transaction Processing component can wrap multi-step EMI postings in one atomic transaction against one data store. A microservices split would scatter this state across independently owned service databases, turning a single ACID transaction into a much harder distributed (eventually-consistent) transaction.

Reason 2 — Manageable Scope with Clear Separation of Concerns

The platform's core logic — credit-risk scoring, repayment scheduling, and default-warning generation — is closely related, moderate in scale, and evolves together rather than needing independent deployment or independent scaling. Layering keeps the Presentation, Business, and Data concerns cleanly separated (easing testing and maintenance) without the operational overhead of running and coordinating many independently deployed services, which the assigned scenario does not warrant.

Security Advantage

Because the Data Layer is only reachable through the Business Layer's components (Risk Assessment and Transaction Processing), the Database is never exposed directly to the Presentation Layer or the end user. All borrower financial data and credit-risk profiles are mediated by business-rule components that can enforce authentication, authorisation, and input validation before any query reaches the database — reducing the attack surface for injection or unauthorised data-access attempts.

Performance Benefit

A single, co-located Database component avoids the network round-trips and distributed-transaction coordination overhead that a microservices-per-data-store design would introduce for EMI processing. This keeps repayment and risk-scoring transactions inside one process/database boundary, helping the system meet the low-latency benchmarks required under simulated peak load in NFR-001.

UML Component Diagram

Micro-Lending & Peer Credit Risk Assessor — UML Component Diagram

Architectural Style: Layered Architecture (Presentation / Business / Data)

